# Negativity Sells: A Sentiment Analysis of Huff-Post Political Articles

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## 1. Introduction

**Background and Motivation:**

In today's digital-first media environment, the competition for reader attention is not just intense - it is relentless. News organizations operate within an ecosystem shaped by social media feeds, push notifications, and algorithm-driven content rankings, where headlines function as the primary gateway to engagement. A headline is no longer simply a summary of a story; it is a strategic tool designed to stop scrolling thumbs, generate clicks, and convert fleeting attention into measurable traffic. For many outlets, this traffic directly translates into advertising revenue and platform visibility, creating strong incentives to optimize headlines for maximum engagement.

Empirical evidence underscores just how powerful headlines have become. According to a study by the Pew Research Centre, 63% of U.S. adults admit to sharing news articles after reading only the headline, without engaging with the full content of the story. Among Millennials and Gen Z, this figure exceeds 70% [1]. These statistics suggest that for a majority of readers, particularly younger audiences, the headline is often the only information consumed before forming opinions or amplifying content through sharing. In such an environment, the informational weight of articles is disproportionately placed on their headlines rather than the actual content. A few carefully chosen words can shape interpretation, emotional response, and even broader public discourse.
This reality raises important concerns about the incentives facing news organizations. If headlines are the primary drivers of clicks and shares, outlets may be tempted to craft language that maximizes emotional engagement. Psychological research consistently shows that individuals exhibit stronger and faster reactions to negative stimuli compared to neutral or positive ones - a phenomenon commonly described as negativity bias [2]. Negative information tends to capture attention more effectively, linger longer in memory, and provoke stronger emotional responses. From a behavioural perspective, emotionally charged or negative headlines may therefore be particularly effective at drawing readers in.

As news consumption increasingly migrates to digital platforms, where metrics such as click-through rates, engagement time, and social shares are constantly tracked and optimized, the strategic use of negative language becomes an especially salient issue. If negativity systematically increases engagement, it may be rational, from a business standpoint, for outlets to emphasize conflict, crisis, and alarm in their headlines. However, this dynamic carries broader societal implications. Overexposure to negative framing may shape public perceptions of risk, political polarization, and societal stability. It may also contribute to fatigue, anxiety, or declining trust in media institutions.

Understanding whether and to what extent emotional framing is embedded in digital headlines is therefore not merely an academic question. It sits at the intersection of media economics, psychology, and democratic discourse. In modern digital news environments, headlines function as the primary gateway to engagement, shaping whether readers choose to click, read, or share an article. As competition for attention intensifies, news organizations may increasingly rely on emotionally charged language to capture reader interest.

Examining patterns in headline language can therefore provide insight into how competitive pressures may be influencing journalistic practices. If certain emotional tones or linguistic strategies appear systematically across categories of news content, this may suggest that headlines are crafted not only to summarize information but also to maximize engagement. By analysing sentiment, emotional language, and linguistic features such as intensifiers across a large dataset of news headlines, this study aims to better understand how emotional framing differs across types of news content and how headlines may shape the way audiences interpret and interact with information in the modern media landscape.

**Research Goal:**
The primary goal of this study is to examine how emotional framing differs across types of online news content. Specifically, the analysis investigates whether the emotional tone and linguistic structure of news headlines vary systematically across different categories of articles in the Huffington Post dataset. The study focuses on four categories - U.S. News, World News, Entertainment, and Travel - which represent distinct forms of journalism with differing subject matter and audience expectations.

U.S. News and World News primarily covers politics, public policy, global events, and other major societal developments. These topics often involve conflict, crisis, or controversy, which may naturally influence the emotional tone of headlines. In contrast, Entertainment and Travel articles tend to focus on lifestyle-oriented topics such as celebrity culture, leisure activities, and cultural trends. These forms of content may be more likely to emphasize positive or engaging language rather than urgency or alarm. By comparing headlines across these categories, the study aims to determine whether emotional language differs systematically depending on the type of news being presented.

Beyond basic sentiment differences, the study also examines additional dimensions of emotional framing. First, it compares the sentiment of headlines with the sentiment of their corresponding article summaries in order to evaluate whether headlines systematically differ in tone relative to the article content they represent. Second, it analyses linguistic features such as emotional words, intensifiers, and downtoners that may amplify or soften emotional tone in headlines. Together, these analyses provide a more comprehensive understanding of how language is used to frame news stories and capture reader attention in digital media environments.

The central research question guiding this study is: **Do headlines in hard news categories systematically contain more negative and emotionally intense language than headlines in lifestyle-oriented news categories?** Addressing this question can help illuminate whether emotional tone in headlines is primarily driven by the underlying subject matter or whether broader engagement incentives influence how news is framed. By examining sentiment, emotional language, and linguistic intensity across categories, this study seeks to provide a clearer picture of how emotional framing operates in modern digital journalism.

**Hypothesis:**
The primary hypothesis of this study is that headlines in hard news categories, specifically U.S. News and World News, will exhibit more negative sentiment than headlines in lifestyle-oriented categories such as Entertainment and Travel. This expectation is grounded in both the subject matter typically covered in these categories and the well-documented psychological phenomenon of negativity bias. Hard news frequently involves topics such as political conflict, economic instability, crime, and global crises, which are inherently associated with negative language and emotionally charged framing. In contrast, lifestyle-oriented categories

often focus on leisure, culture, and personal experiences, which may naturally lend themselves to more neutral or positive language.

In addition to this primary hypothesis, the study explores two related expectations regarding emotional framing in headlines. First, headlines may display a stronger emotional tone than the article summaries they accompany, reflecting the role of headlines as tools for attracting reader attention. Second, lifestyle-oriented headlines may make greater use of linguistic intensifiers that amplify emotional tone, while hard news headlines may rely more heavily on direct negative framing associated with the events being reported. Together, these expectations allow the analysis to evaluate not only differences in sentiment across news categories but also broader patterns in how emotional language is structured in digital headlines.

## 2. Data

The dataset used in this study was compiled by machine learning researcher Rishabh Misra and contains 209,527 articles published by the Huffington Post between 2012 and 2022. The dataset was originally created to support machine learning and natural language processing research and includes several structured variables for each article, including the article URL, headline, category, short description, author, and publication date. Because the dataset contains both textual information and categorical labels, it is well-suited for analysing patterns in language across different types of news content.

For the purposes of this analysis, the study focuses on four categories within the dataset: U.S. News (1,377 articles), World News (3,299 articles), Entertainment (17,362 articles), and Travel (9,900 articles). These categories were selected because they represent distinct forms of journalism that differ in both subject matter and typical emotional framing. U.S. News and World News primarily covers political developments, global events, economic issues, and other major societal topics. In contrast, Entertainment and Travel focus on lifestyle-oriented content such as celebrity culture, leisure activities, and cultural trends. Restricting the analysis to these categories allows for a structured comparison between hard news reporting and more lifestyle-oriented coverage.

Within the dataset, the headline field serves as the primary unit of analysis, as headlines represent the most visible and influential textual component of digital news consumption. In online environments where readers often skim content or share articles based solely on the headline, the language used in this field plays a critical role in shaping perception and engagement. The short description field, which provides a brief summary of each article, is also used in the analysis to compare headline sentiment with the tone of the underlying article content.

Although the dataset spans a ten-year period, the majority of articles were published between 2012 and 2018, with a smaller portion appearing between 2018 and 2022. The large size of the dataset and its structured format allow for systematic and replicable analysis of linguistic patterns across news categories.

## 3. Methodology:

To examine patterns in emotional framing across news headlines, this study applies several natural language processing and statistical techniques. The analysis focuses on measuring sentiment, identifying emotional language, comparing headline tone across categories, and evaluating differences between headlines and article

summaries. All analyses were conducted using the R programming language and several packages designed for text analysis, including *tidytext*, *dplyr*, and *tidyr*.

Before analysis, the headline text was pre-processed using standard text-cleaning procedures. Headlines were converted to lowercase and tokenized into individual words. Tokenization allows the text to be broken into smaller components that can be systematically compared against established sentiment and emotion lexicons. These processed tokens were then used to compute sentiment scores, emotional word counts, and other linguistic features.

**Sentiment Analysis**

To measure the overall tone of headlines, the study uses a lexicon-based sentiment analysis approach. Specifically, the Bing sentiment lexicon was applied to classify words appearing in headlines as either positive or negative. After tokenization, each word in a headline was matched with the lexicon to determine whether it contributed positive or negative sentiment.

For each headline, a sentiment score was calculated as the difference between the number of positive and negative words appearing in the headline: Sentiment Score = Positive Words − Negative Words.
A higher score indicates a more positive headline, while a lower score indicates greater negativity. Headlines were then categorized as positive, negative, or neutral based on their sentiment score. These scores were used to examine how sentiment distributions differ across news categories.

**Emotion Lexicon Analysis**

While positive and negative sentiment provide a general measure of tone, it does not capture the specific emotional signals present in language. To examine emotional framing more closely, the study also applies the NRC emotion lexicon, which classifies words into eight emotional categories: anger, fear, sadness, joy, anticipation, trust, surprise, and disgust.

Each headline was tokenized and matched with the NRC lexicon to identify words associated with these emotions. The number of words corresponding to each emotional category was then counted for every headline. These counts allow for a more detailed comparison of how different emotions appear across hard news and lifestyle-oriented categories.

To evaluate whether emotional word usage differs systematically across categories, linear regression models were estimated with the counts of specific emotion words as the dependent variables and news category as the independent variable.

**Hard News vs. Lifestyle News Comparison**

To simplify comparisons across categories, the four selected sections were grouped into two broader categories. U.S. News and World News were classified as Hard News, while Entertainment and Travel were classified as General News. This grouping allows for direct statistical comparisons between traditional news reporting and lifestyle-oriented content.

Linear regression models were used to evaluate whether headline sentiment differs significantly between these groups. In these models, headline sentiment scores served as the dependent variable, and news category (Hard

News vs. General News) served as the explanatory variable. This approach allows the analysis to estimate the magnitude and statistical significance of differences in headline tone across categories.

**Headline vs. Article Summary Comparison**

To evaluate whether headlines differ in tone relative to the content they represent, sentiment scores were also computed for the short descriptions associated with each article. The same sentiment scoring procedure used for headlines was applied to the summary text.

Because each headline is directly paired with its corresponding article summary, the study used paired t-tests to compare the sentiment scores of headlines and summaries within the same article. This within-article comparison allows the analysis to determine whether headlines systematically differ in emotional tone from the underlying article descriptions.

**Linguistic Intensifiers and Downtoners**

In addition to sentiment and emotion analysis, the study also examines linguistic modifiers that influence emotional intensity. Intensifiers are words that amplify emotional tone, such as "dramatically," "remarkably," or "extraordinarily." Conversely, downtoners are words that soften or reduce emotional intensity.

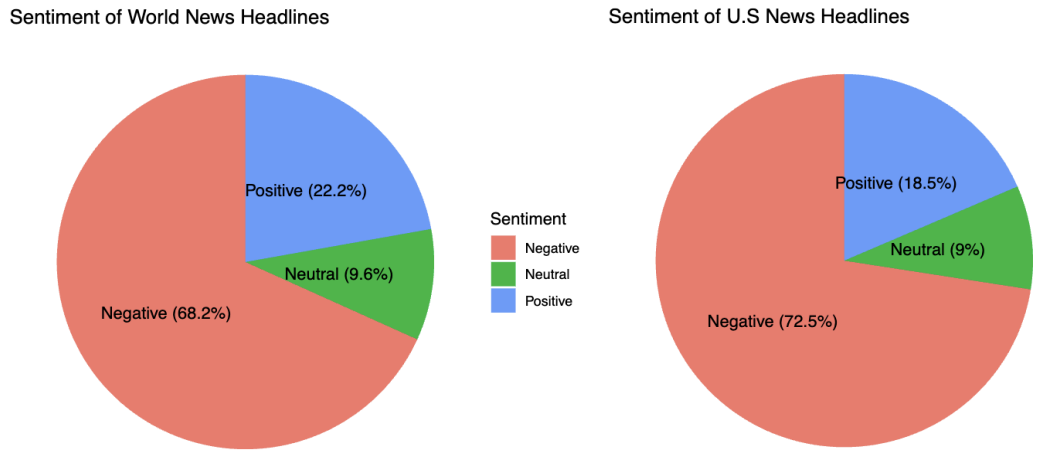
Using valence shifter dictionaries from the *lexicon* package, along with additional manually defined intensifiers, the analysis identifies headlines containing these modifiers. The number of intensifiers and downtoners appearing in each headline was counted and compared across news categories. Linear regression models were then used to test whether the frequency of these linguistic features differs between hard news and lifestyle-oriented headlines.

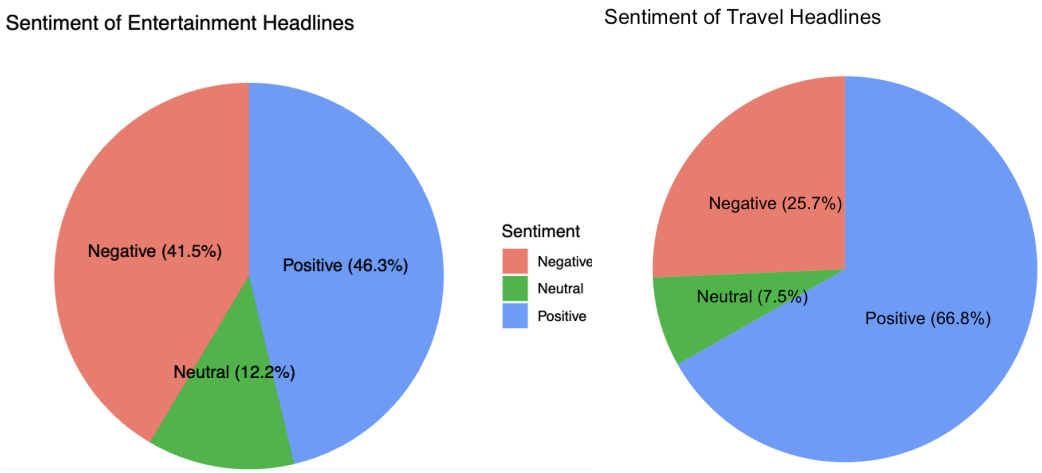
## 4. <u>Results</u>

**Sentiment Distribution Across News Categories**

To evaluate differences in emotional tone across news categories, sentiment scores were first computed for each headline using the Bing sentiment lexicon. Headlines were then classified as positive, negative, or neutral based on the difference between the number of positive and negative words appearing in each headline.

Figure 1 illustrates the distribution of headline sentiment across the four categories examined in this study: U.S. News, World News, Entertainment, and Travel.





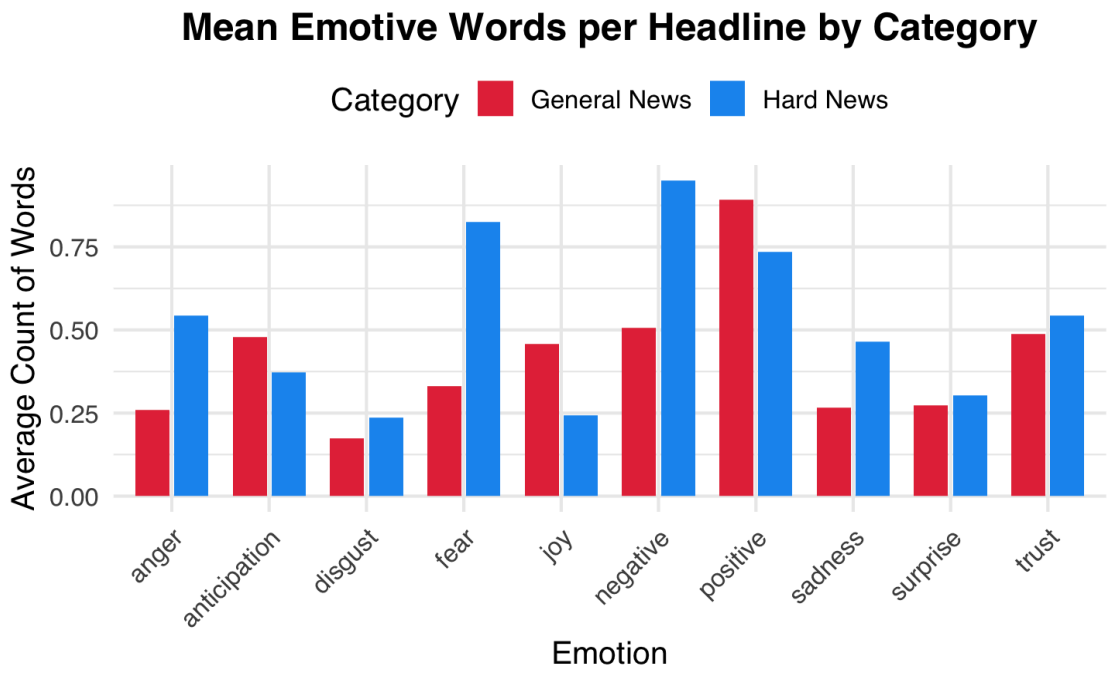

The results reveal clear differences across categories. In the World News category, approximately 68.2% of headlines are classified as negative, compared to 22.2% positive and 9.6% neutral. A similar pattern appears in U.S. News, where 72.5% of headlines are negative, 18.5% positive, and 9% neutral. In both categories, negative headlines substantially outnumber positive ones, suggesting that hard news reporting relies heavily on negative framing. In contrast, the lifestyle-oriented categories display markedly different sentiment patterns. Entertainment headlines show a more balanced distribution, with 46.3% positive, 41.5% negative, and 12.2% neutral. Meanwhile, Travel headlines are predominantly positive, with 66.8% positive, 25.7% negative, and 7.5% neutral. These descriptive results provide initial evidence that emotional tone differs systematically depending on the type of news content being reported.

To formally test whether sentiment differs between types of news, a linear regression model was created comparing average headline sentiment scores between Hard news categories (U.S. News and World News) and lifestyle-oriented categories (Entertainment and Travel). The results indicate a significant difference in sentiment between the two groups. Headlines in lifestyle-oriented news have an average sentiment score of 0.178, while headlines in hard news are approximately 0.905 points lower on average ($\beta = -0.905$, $p < 8.02e\text{-}300$). This result confirms that hard news headlines are substantially more negative than those appearing in general news categories.

**Emotional Language Across Categories**

While overall sentiment provides a broad measure of tone, it does not capture the specific emotional signals embedded in headlines. To examine this dimension more closely, the analysis applied the NRC emotion lexicon, which categorizes words into emotional groups such as anger, fear, sadness, and joy.

Figure 2 compares the average number of emotion-related words appearing in headlines across hard news and general news categories.



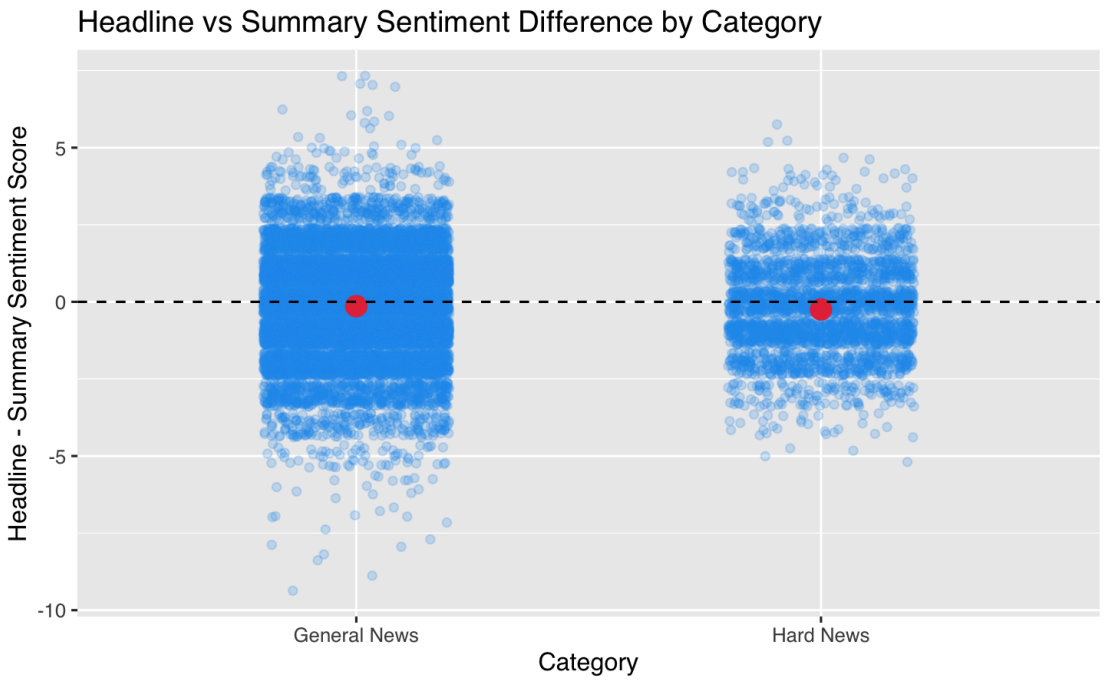

The results show that hard news headlines contain higher frequencies of words associated with negative emotions, particularly fear, anger, and sadness. These emotions are commonly linked to topics such as political conflict, crime, and global crises, which frequently appear in U.S. News and World News coverage. In contrast, general news headlines contain relatively more words associated with positive emotions, including joy and anticipation, reflecting the more lifestyle-oriented nature of entertainment and travel reporting. This pattern suggests that emotional tone in headlines is closely tied to the underlying subject matter of the news being reported. Hard news topics often involve uncertainty, risk, or conflict, which may naturally lead to stronger expressions of negative emotion. Lifestyle-oriented topics, by contrast, tend to emphasize enjoyment, cultural experiences, and personal interests, which may contribute to a more positive emotional tone in the language used to frame these stories.

To formally test the differences in the emotive words used in headlines between types of news, a linear regression model was created comparing the mean number of emotive words per emotional category between hard news categories and lifestyle-oriented categories. The results indicate a significant difference in emotional tone of headlines in each group. Hard news headlines were on average found to contain more anger ($\beta = 0.283$, $p < 1.447974e\text{-}193$), slightly more disgust ($\beta = 0.062$, $p < 4.18e\text{-}17$), significantly more fear ($\beta = 0.493$ , $p < 2e\text{-}16$), and more sadness ( $\beta = 0.198$, $p < 1.02e\text{-}101$). However they were also found to have slightly more surprise ( $\beta = 0.031$, $p < 0.000305$), and slightly more trust ( $\beta = 0.055$, $p < 1.28e\text{-}06$). And they also had less anticipation ($\beta = -0.1061952$, $p < 8.49e\text{-}23$) and joy ( $\beta = 0.2152641$, $p < 3.614e\text{-}90$). These results help show that headlines in hard news categories tend to have more negative emotional connections than lifestyle-oriented categories.

**Headline Sentiment vs Article Summary Sentiment**

To determine whether headlines differ in tone relative to the articles they represent, sentiment scores were also calculated for the short descriptions associated with each article. Because each headline is paired with a corresponding summary, this analysis allows for a direct comparison between headline sentiment and article sentiment.

Figure 3 displays the distribution of the difference between headline sentiment and summary sentiment across the two news groups.



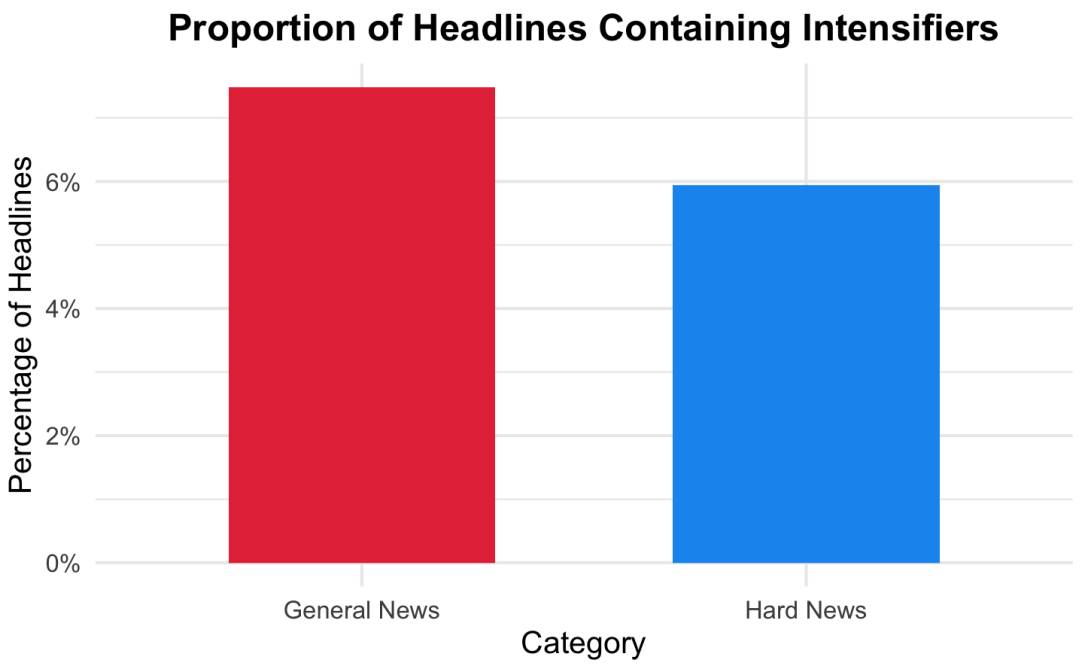

Across both groups, the results indicate that headlines tend to be slightly more negative than their corresponding article summaries. This pattern suggests that headlines may emphasize stronger emotional language than the article content itself, potentially reflecting the role of headlines as tools for capturing reader attention in competitive digital media environments. Because headlines serve as the primary point of contact between readers and news content, they may be strategically framed to evoke stronger emotional reactions that encourage clicks and engagement. As a result, the tone of a headline may not always fully reflect the tone of the article it represents, instead highlighting the most dramatic or emotionally charged elements of a story. This finding provides additional evidence that headline language may be shaped not only by the underlying content of the article but also by the broader incentives of digital media platforms.

To formally test the differences between headlines and articles summaries between types of news, a linear regression model was created comparing the mean sentiment of headlines between hard news categories and lifestyle-oriented categories. Although both headlines tended to be more negative than the summaries on average, hard news headlines were more negative to a larger extent than lifestyle news ($\beta = -0.10719$, $p < 0.00043$).

**Linguistic Intensifiers and Downtoners**

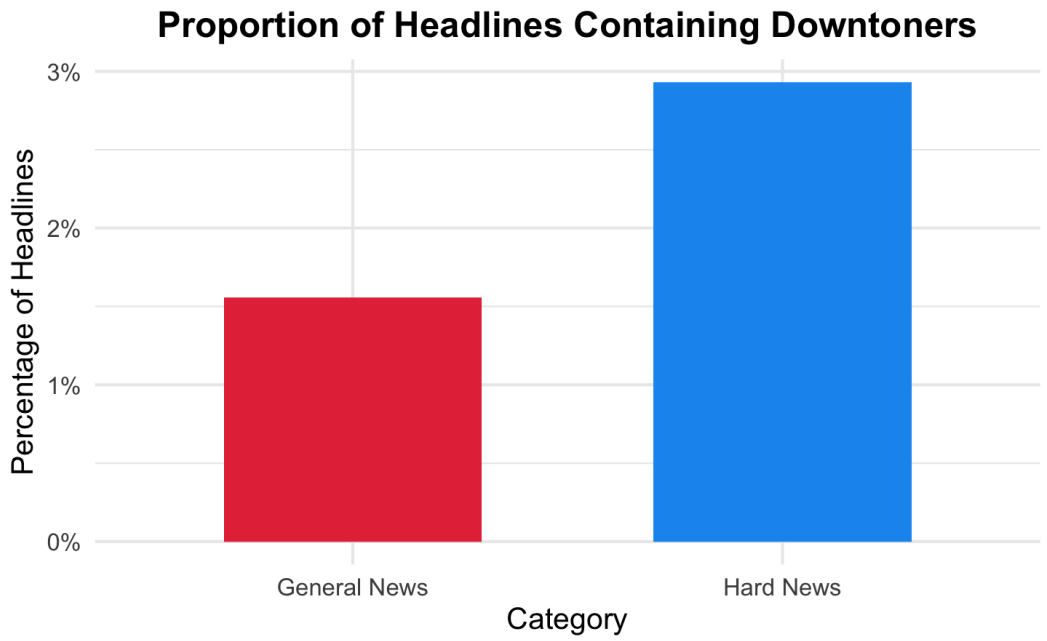
In addition to sentiment and emotional word counts, the analysis also examines linguistic modifiers that influence emotional intensity. Intensifiers are words that amplify the strength of statements, such as “dramatically” or “remarkably,” while downtoners soften emotional language.

Figure 4 compares the proportion of headlines containing intensifiers across hard news and general news categories.




The results show that general news headlines contain a slightly higher proportion of intensifiers than hard news headlines. This pattern suggests that lifestyle-oriented content may rely more heavily on exaggerated or attention-grabbing language in order to engage readers and emphasize excitement or novelty. Topics such as entertainment and travel often highlight experiences, events, or cultural trends, which may lend themselves to more expressive or amplified language in headlines. In contrast, hard news reporting typically focuses on political developments, economic issues, or global events, where exaggerated wording may be less common.

Figure 5 presents the proportion of headlines containing downtoners across the same categories.




Although downtoners appear relatively infrequently overall, they are slightly more common in hard news headlines. This difference may reflect a more cautious or measured tone when reporting serious topics such as political conflict, crime, or international crises. Journalistic conventions in these areas may encourage more

restrained language in order to maintain credibility and avoid overstating uncertain developments. Together, these patterns suggest that different types of news content may rely on distinct linguistic strategies when framing headlines, with lifestyle-oriented categories emphasizing amplification and hard news categories showing slightly greater use of moderating language.

To formally test the differences in the use of intensifiers and downtoners headlines between types of news, a linear regression model was created comparing the mean number of intensifiers and downtoners per headline between hard news categories and lifestyle-oriented categories. It was found that on average hard news headlines use slightly less intensifiers than lifestyle-oriented headlines ($\beta = -0.016259$, $p < 0.0002$) and use slightly more downtoners ($\beta = 0.0133790$, $p < 3.62e-10$).

## 5. Discussion

The results of this study provide consistent evidence that the emotional tone of headlines varies systematically across different types of news content. Across multiple analyses, including sentiment distributions, emotion lexicon comparisons, headline–summary sentiment differences, and linguistic intensity features, clear distinctions emerge between hard news categories and lifestyle-oriented categories. Headlines in U.S. News and World News are substantially more negative and contain higher frequencies of emotional signals associated with fear, anger, and sadness, while Entertainment and Travel headlines display more positive sentiment and greater use of expressive language.

These findings align closely with the concept of negativity bias, which suggests that individuals tend to respond more strongly to negative information than to neutral or positive stimuli. Psychological research has consistently shown that negative events capture attention more quickly, are processed more deeply, and are remembered more vividly than positive events. As a result, negative information often has a disproportionate influence on perception and decision-making. In the context of news consumption, this tendency means that headlines emphasizing conflict, crisis, or danger may be more likely to draw attention than headlines framed in neutral or positive terms. Because negative information tends to capture attention more effectively and provoke stronger emotional reactions, it may be particularly effective in headline writing. Readers scrolling through social media feeds or news aggregators are often exposed to large volumes of competing information, and headlines must compete for a limited amount of attention. In such environments, emotionally charged or negative language may help a headline stand out and encourage readers to engage with the content. In digital media environments where reader engagement directly influences advertising revenue, platform visibility, and algorithmic rankings, the incentives to capture attention become even stronger. Articles that generate higher levels of clicks, shares, or reading time are more likely to be promoted by recommendation algorithms and distributed more widely across social platforms. As a result, emphasizing negative or emotionally intense language in headlines may increase the likelihood that readers click on or share an article, even if the tone of the underlying article content is more balanced. Over time, these incentives may subtly shape how news organizations frame headlines, encouraging language that maximizes engagement within increasingly competitive digital information ecosystems.

The comparison between headlines and article summaries further supports this interpretation. The analysis shows that headlines tend to be slightly more negative than the summaries of the articles they accompany. This pattern suggests that headline language may selectively highlight the most dramatic or emotionally charged aspects of a story in order to capture the reader's attention. Because headlines function as the primary entry

point into an article, they often prioritize immediacy and emotional impact over complete contextual balance. While the underlying articles may present more nuanced or balanced descriptions of events, the headline typically serves as a condensed and highly visible representation of the story. In practice, this means that headline writers may emphasize elements such as conflict, controversy, or urgency-features that are more likely to attract attention in fast-paced digital environments. As readers increasingly encounter news through social media feeds, search results, and mobile notifications, the headline often becomes the main determinant of whether a user engages with an article. This dynamic may also help explain why the sentiment difference between headlines and summaries appears consistently across categories. Even when the overall article content is relatively neutral or balanced, the headline may focus on the most striking or emotionally salient aspect of the story. Over time, this tendency can contribute to a broader pattern in which headlines appear systematically more negative or emotionally intense than the content they summarize, reflecting the growing importance of engagement-driven headline writing in modern digital journalism.

Differences in linguistic strategies across categories also provide insight into how emotional framing varies by topic. Lifestyle-oriented categories such as Entertainment and Travel appear to rely more heavily on intensifiers, which amplify emotional tone and create a sense of excitement or novelty. Words that heighten emphasis can make headlines feel more engaging or dramatic, which may be particularly effective for content designed to entertain readers or highlight appealing experiences. Because articles in these categories often focus on leisure, culture, or personal interests, headline writers may use more expressive language to capture attention and generate enthusiasm. Hard news headlines, by contrast, contain slightly more downtoners, suggesting a somewhat more restrained or cautious tone when reporting serious topics. News related to politics, crime, or global events often involves uncertainty, evolving information, or sensitive issues, which may encourage journalists to adopt more measured language. Using moderating language can help maintain credibility and avoid overstating developments before all details are fully confirmed. These differences indicate that headline language may be shaped not only by subject matter but also by the expectations and conventions associated with different forms of journalism, reflecting distinct stylistic approaches across news categories.

Taken together, the findings suggest that emotional framing in digital headlines reflects a combination of subject-matter differences and engagement-driven incentives within modern media environments. While negative language in hard news may partly reflect the nature of the events being reported, such as political conflict, crime, or global crises, the patterns observed in the analysis indicate that editorial choices may also play an important role in shaping how stories are presented to readers. In particular, the tendency for headlines to emphasize stronger emotional language than the article summaries they accompany suggests that headline framing may be designed to highlight the most dramatic or attention-grabbing aspects of a story. These dynamics are especially relevant in digital media ecosystems where headlines compete for visibility across crowded social media feeds, search results, and news aggregation platforms. In such environments, the ability of a headline to quickly capture reader attention can directly influence whether an article is clicked, shared, or widely circulated. As a result, emotional tone may function not only as a reflection of the content itself but also as a strategic tool for increasing engagement. Together, these findings highlight how both the nature of news events and the competitive pressures of digital journalism can shape the language used in headlines.

## 6. Conclusion

This study set out to examine how emotional framing varies across different types of digital news headlines. Using a large dataset of 209,527 Huffington Post articles, the analysis focused on four news categories-U.S.

News, World News, Entertainment, and Travel - in order to compare the emotional tone of headlines across hard news and lifestyle-oriented content. By applying a combination of sentiment analysis, emotion lexicon analysis, headline–summary sentiment comparisons, and linguistic feature analysis, the study sought to evaluate whether emotional language in headlines differs systematically depending on the subject matter being reported.

The results reveal clear and consistent differences in the emotional tone of headlines across categories. Headlines in U.S. News and World News are overwhelmingly negative in sentiment, with a substantial majority classified as negative according to the Bing sentiment lexicon. These categories also contain higher frequencies of words associated with emotions such as fear, anger, and sadness. These emotional patterns are closely tied to the nature of the topics commonly covered in hard news reporting, which frequently involves political conflict, economic instability, crime, and global crises. In contrast, lifestyle-oriented categories such as Entertainment and Travel exhibit markedly different sentiment patterns. Entertainment headlines appear more balanced between positive and negative sentiment, while Travel headlines are predominantly positive and contain more language associated with emotions such as joy and anticipation. These differences suggest that the emotional tone of headlines reflects the broader thematic orientation of the content being presented.

Beyond simple sentiment differences, the analysis also reveals important patterns in how headlines frame stories relative to the underlying article content. By comparing the sentiment of headlines with the sentiment of their corresponding article summaries, the study finds that headlines tend to be slightly more negative than the summaries they accompany. This finding suggests that headlines may selectively emphasize the most dramatic or emotionally charged aspects of a story. While article summaries often provide a more balanced description of events, the headline functions as a condensed representation designed to capture reader's attention. In digital media environments where readers encounter large volumes of information through social media feeds, search engines, and mobile notifications, headlines play a crucial role in determining whether an article is noticed or engaged with. The analysis of linguistic features further illustrates how headline framing differs across news categories. Lifestyle-oriented headlines appear to rely more heavily on intensifiers, which amplify emotional tone and create a sense of excitement or novelty. Such language may be particularly effective for content designed to entertain readers or highlight appealing experiences. Hard news headlines, by contrast, contain slightly more downtoners, suggesting a somewhat more cautious or measured tone when reporting serious topics. This difference may reflect journalistic conventions that encourage more restrained language when discussing uncertain or sensitive developments in political or global news.

These findings suggest that emotional framing in digital headlines is shaped by a combination of subject-matter differences and engagement-driven incentives within modern media environments. On one hand, the prevalence of negative sentiment in hard news headlines reflects the inherently serious or conflict-oriented nature of many news events. On the other hand, the tendency for headlines to emphasize stronger emotional language than article summaries indicates that editorial decisions about headline wording may also be influenced by the competitive pressures of digital journalism. In online environments where headlines compete for visibility and reader attention, emotional language can function as a powerful tool for increasing engagement. Overall, the results highlight the important role that headlines play in shaping how news is presented and perceived. Because headlines often represent the first, and sometimes only, interaction that readers have with a news story, the language used in these brief pieces of text can significantly influence how information is interpreted and shared. Understanding the emotional patterns embedded in headlines, therefore, provides valuable insight into the broader dynamics of digital news production and consumption. By examining how sentiment, emotional

language, and linguistic intensity vary across categories of news content, this study contributes to a deeper understanding of how modern media environments shape the framing of information presented to audiences.

## 7. Limitations

While this study provides useful insights into emotional framing in digital news headlines, several limitations should be considered when interpreting the results.

First, the dataset used in this analysis was compiled by an external researcher rather than directly by HuffPost. As a result, the category labels and article summaries included in the dataset may not perfectly reflect the publisher's internal classification system. Although the categories appear broadly consistent with HuffPost's editorial structure, some articles may be misclassified or overlap across multiple topical areas. For example, an article categorized as "U.S. News" could include elements of opinion, commentary, or political analysis, which may influence the measured sentiment of the headline. Any inconsistencies in categorization could therefore affect comparisons across news sections. Also, the dataset is heavily concentrated in articles published between 2012 and 2018, with only a small proportion of observations from more recent years. Because digital media environments evolve rapidly, headline writing strategies and audience engagement patterns may have changed since the earlier portion of the dataset was collected. In particular, the growth of social media platforms, algorithmic news feeds, and mobile-first news consumption may have further influenced how headlines are written and optimized for engagement. As a result, the patterns observed in this study may not fully capture the most recent developments in digital journalism.

Moreover, another limitation relates to the methodology used to measure sentiment and emotional tone. The study relies on lexicon-based sentiment analysis techniques, which classify words according to predefined dictionaries of positive, negative, or emotion-related terms. While this approach allows for systematic and replicable analysis across large volumes of text, it does not fully capture contextual nuances in language. Words that are classified as negative or positive in isolation may carry different meanings depending on their context within a sentence. Additionally, lexicon-based methods may struggle to detect sarcasm, irony, or more complex forms of linguistic framing. Finally, the analysis focuses exclusively on the textual content of headlines and article summaries. Other elements that influence reader engagement, such as images, article placement, publication timing, or social media presentation, are not captured in the dataset. Because digital news consumption often involves multiple visual and contextual cues, the emotional framing of headlines represents only one component of the broader mechanisms that shape reader attention and engagement.

Despite these limitations, the dataset remains large and well-structured, allowing for systematic comparison of linguistic patterns across multiple news categories. The results should therefore be interpreted as evidence of general tendencies in headline framing rather than precise measurements of editorial intent.

### Scope for Further Work

While this study provides insight into emotional framing patterns in digital news headlines, several opportunities exist for extending the analysis in future research. One potential direction would be to incorporate more advanced natural language processing techniques. The current study relies primarily on lexicon-based sentiment analysis methods, which classify words using predefined dictionaries. Future work could apply machine learning or transformer-based language models to capture more complex linguistic patterns, contextual meanings, and subtle emotional cues that dictionary approaches may miss.

Another avenue for further research would be to expand the dataset to include a broader range of news outlets and publication sources. Because this study focuses solely on articles from the Huffington Post, the findings may reflect editorial styles specific to that publication. Analysing headlines across multiple news organizations-such as traditional newspapers, digital-native outlets, and international publications-could help determine whether the emotional patterns observed in this study are consistent across the broader media landscape. Future research could also examine how headline sentiment relates to actual audience engagement. Metrics such as click-through rates, social media shares, comment activity, or reading time could provide additional insight into whether emotionally intense or negative headlines generate higher levels of reader interaction. By linking linguistic features of headlines to engagement data, researchers could better evaluate the role that emotional framing plays in shaping news consumption behaviour.

Finally, future studies could explore how headline sentiment has evolved over time. Because digital media environments have changed rapidly in recent years, particularly with the rise of algorithm-driven news feeds and social media platforms, longitudinal analysis could reveal whether emotional framing strategies in headlines are becoming more intense or polarized over time. Such work would provide a deeper understanding of how journalistic practices adapt to changing technological and economic conditions within modern media ecosystems.

**Sources**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